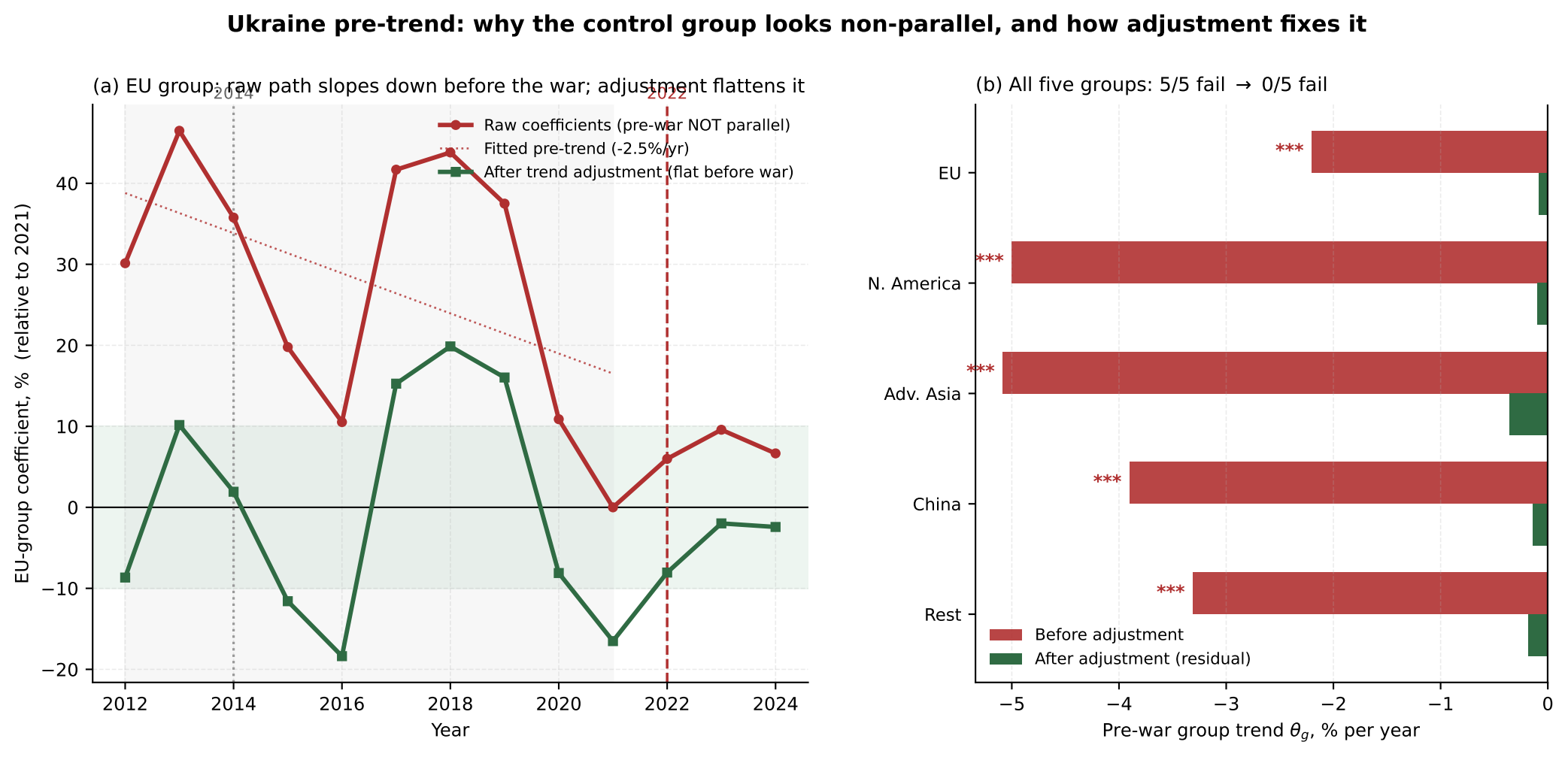
Ukraine pre-trend: why the control group looks non-parallel, and how adjustment fixes it
(a) EU group: raw path slopes down before the war; adjustment flattens it
2014
2022
Raw coefficients (pre-war NOT parallel)
Fitted pre-trend (-2.5%/yr)
After trend adjustment (flat before war)
EU-group coefficient, % (relative to 2021)
40
30
20
10
0
−10
−20
2012
2014
2016
2018
2020
2022
2024
Year
(b) All five groups: 5/5 fail → 0/5 fail
EU
N. America
Adv. Asia
China
Rest
***
Before adjustment
After adjustment (residual)
−5
−4
−3
−2
−1
0
Pre-war group trend θ_g, % per year